

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	V
Subject Code & Name	ICS1512 & Machine Learning Algorithms Laboratory		
Academic year	2025-2026 (Odd)	Batch:2023-2028	Due date:

Experiment 3: Email Spam or Ham Classification using Naive Bayes, KNN, and SVM

1 Aim:

To develop and evaluate supervised machine learning models using K-Nearest Neighbors (KNN) and Naive Bayes (Bernoulli, Multinomial, Gaussian) classifiers. To analyze and compare the performance of these models on a given dataset using accuracy and other evaluation metrics.

2 Libraries used:

- Numpy
- Pandas
- Matplotlib
- Scikit-learn
- Seaborn

3 Objective:

- To implement KNN and Naive Bayes algorithms for classification, including preprocessing, training, and testing using appropriate techniques such as cross-validation.
- To assess the models using metrics like accuracy, precision, recall, F1-score, confusion matrix, ROC curve, and AUC, and to present a comparative analysis.

4 Naive Bayes Code:

naive

August 5, 2025

0.0.1 Importing Required Libraries

```
[19]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler # or MinMaxScaler, if used
from sklearn.preprocessing import LabelEncoder # if categorical labels
from sklearn.naive_bayes import BernoulliNB, MultinomialNB, GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, \
    classification_report
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.model_selection import cross_val_score, KFold
```

0.0.2 Loading dataset

```
[2]: df = pd.read_csv('spambase.csv')
```

0.0.3 Basic info

```
[3]: print("Dataset Info:\n", df.info())
print("\nFirst 5 rows:\n", df.head())
print("\nMissing Values:\n", df.isnull().sum())
```

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 4601 entries, 0 to 4600

Data columns (total 58 columns):

#	Column	Non-Null Count	Dtype
0	word_freq_make	4601 non-null	float64
1	word_freq_address	4601 non-null	float64
2	word_freq_all	4601 non-null	float64
3	word_freq_3d	4601 non-null	float64
4	word_freq_our	4601 non-null	float64
5	word_freq_over	4601 non-null	float64
6	word_freq_remove	4601 non-null	float64

7	word_freq_internet	4601	non-null	float64
8	word_freq_order	4601	non-null	float64
9	word_freq_mail	4601	non-null	float64
10	word_freq_receive	4601	non-null	float64
11	word_freq_will	4601	non-null	float64
12	word_freq_people	4601	non-null	float64
13	word_freq_report	4601	non-null	float64
14	word_freq_addresses	4601	non-null	float64
15	word_freq_free	4601	non-null	float64
16	word_freq_business	4601	non-null	float64
17	word_freq_email	4601	non-null	float64
18	word_freq_you	4601	non-null	float64
19	word_freq_credit	4601	non-null	float64
20	word_freq_your	4601	non-null	float64
21	word_freq_font	4601	non-null	float64
22	word_freq_000	4601	non-null	float64
23	word_freq_money	4601	non-null	float64
24	word_freq_hp	4601	non-null	float64
25	word_freq_hpl	4601	non-null	float64
26	word_freq_george	4601	non-null	float64
27	word_freq_650	4601	non-null	float64
28	word_freq_lab	4601	non-null	float64
29	word_freq_labs	4601	non-null	float64
30	word_freq_telnet	4601	non-null	float64
31	word_freq_857	4601	non-null	float64
32	word_freq_data	4601	non-null	float64
33	word_freq_415	4601	non-null	float64
34	word_freq_85	4601	non-null	float64
35	word_freq_technology	4601	non-null	float64
36	word_freq_1999	4601	non-null	float64
37	word_freq_parts	4601	non-null	float64
38	word_freq_pm	4601	non-null	float64
39	word_freq_direct	4601	non-null	float64
40	word_freq_cs	4601	non-null	float64
41	word_freq_meeting	4601	non-null	float64
42	word_freq_original	4601	non-null	float64
43	word_freq_project	4601	non-null	float64
44	word_freq_re	4601	non-null	float64
45	word_freq_edu	4601	non-null	float64
46	word_freq_table	4601	non-null	float64
47	word_freq_conference	4601	non-null	float64
48	char_freq_%3B	4601	non-null	float64
49	char_freq_%28	4601	non-null	float64
50	char_freq_%5B	4601	non-null	float64
51	char_freq_%21	4601	non-null	float64
52	char_freq_%24	4601	non-null	float64
53	char_freq_%23	4601	non-null	float64
54	capital_run_length_average	4601	non-null	float64

```

55 capital_run_length_longest 4601 non-null int64
56 capital_run_length_total    4601 non-null int64
57 class                       4601 non-null int64

```

dtypes: float64(55), int64(3)

memory usage: 2.0 MB

Dataset Info:

None

First 5 rows:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d \
0	0.00	0.64	0.64	0.0
1	0.21	0.28	0.50	0.0
2	0.06	0.00	0.71	0.0
3	0.00	0.00	0.00	0.0
4	0.00	0.00	0.00	0.0

	word_freq_our	word_freq_over	word_freq_remove	word_freq_internet \
0	0.32	0.00	0.00	0.00
1	0.14	0.28	0.21	0.07
2	1.23	0.19	0.19	0.12
3	0.63	0.00	0.31	0.63
4	0.63	0.00	0.31	0.63

	word_freq_order	word_freq_mail	...	char_freq_%3B	char_freq_%28 \
0	0.00	0.00	...	0.00	0.000
1	0.00	0.94	...	0.00	0.132
2	0.64	0.25	...	0.01	0.143
3	0.31	0.63	...	0.00	0.137
4	0.31	0.63	...	0.00	0.135

	char_freq_%5B	char_freq_%21	char_freq_%24	char_freq_%23 \
0	0.0	0.778	0.000	0.000
1	0.0	0.372	0.180	0.048
2	0.0	0.276	0.184	0.010
3	0.0	0.137	0.000	0.000
4	0.0	0.135	0.000	0.000

	capital_run_length_average	capital_run_length_longest \
0	3.756	61
1	5.114	101
2	9.821	485
3	3.537	40
4	3.537	40

	capital_run_length_total	class
0	278	1
1	1028	1
2	2259	1

3	191	1
4	191	1

[5 rows x 58 columns]

Missing Values:

word_freq_make	0
word_freq_address	0
word_freq_all	0
word_freq_3d	0
word_freq_our	0
word_freq_over	0
word_freq_remove	0
word_freq_internet	0
word_freq_order	0
word_freq_mail	0
word_freq_receive	0
word_freq_will	0
word_freq_people	0
word_freq_report	0
word_freq_addresses	0
word_freq_free	0
word_freq_business	0
word_freq_email	0
word_freq_you	0
word_freq_credit	0
word_freq_your	0
word_freq_font	0
word_freq_000	0
word_freq_money	0
word_freq_hp	0
word_freq_hpl	0
word_freq_george	0
word_freq_650	0
word_freq_lab	0
word_freq_labs	0
word_freq_telnet	0
word_freq_857	0
word_freq_data	0
word_freq_415	0
word_freq_85	0
word_freq_technology	0
word_freq_1999	0
word_freq_parts	0
word_freq_pm	0
word_freq_direct	0
word_freq_cs	0
word_freq_meeting	0

```

word_freq_original      0
word_freq_project       0
word_freq_re            0
word_freq_edu           0
word_freq_table         0
word_freq_conference    0
char_freq_%3B           0
char_freq_%28           0
char_freq_%5B           0
char_freq_%21           0
char_freq_%24           0
char_freq_%23           0
capital_run_length_average 0
capital_run_length_longest 0
capital_run_length_total  0
class                   0
dtype: int64

```

0.0.4 Handling missing values

```

[6]: imputer = SimpleImputer(strategy='mean')
     df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)

```

0.0.5 Splitting of feature and target

```

[7]: X = df_imputed.drop('class', axis=1)
     y = df_imputed['class']

```

0.0.6 Checking Distribution

```

[8]: X = df.drop('class', axis=1)  # Assuming 'class' is the target

# Choose a few features to visualize
sample_features = X.columns[:5]  # First 5 features

# Plot histograms with Gaussian curve overlay
plt.figure(figsize=(15, 10))
for i, feature in enumerate(sample_features):
    plt.subplot(3, 2, i + 1)
    data = X[feature]

    # Plot histogram
    count, bins, ignored = plt.hist(data, bins=30, density=True, alpha=0.6,
    color='skyblue', edgecolor='black')

    # Plot normal distribution curve
    '''mu, std = data.mean(), data.std()

```

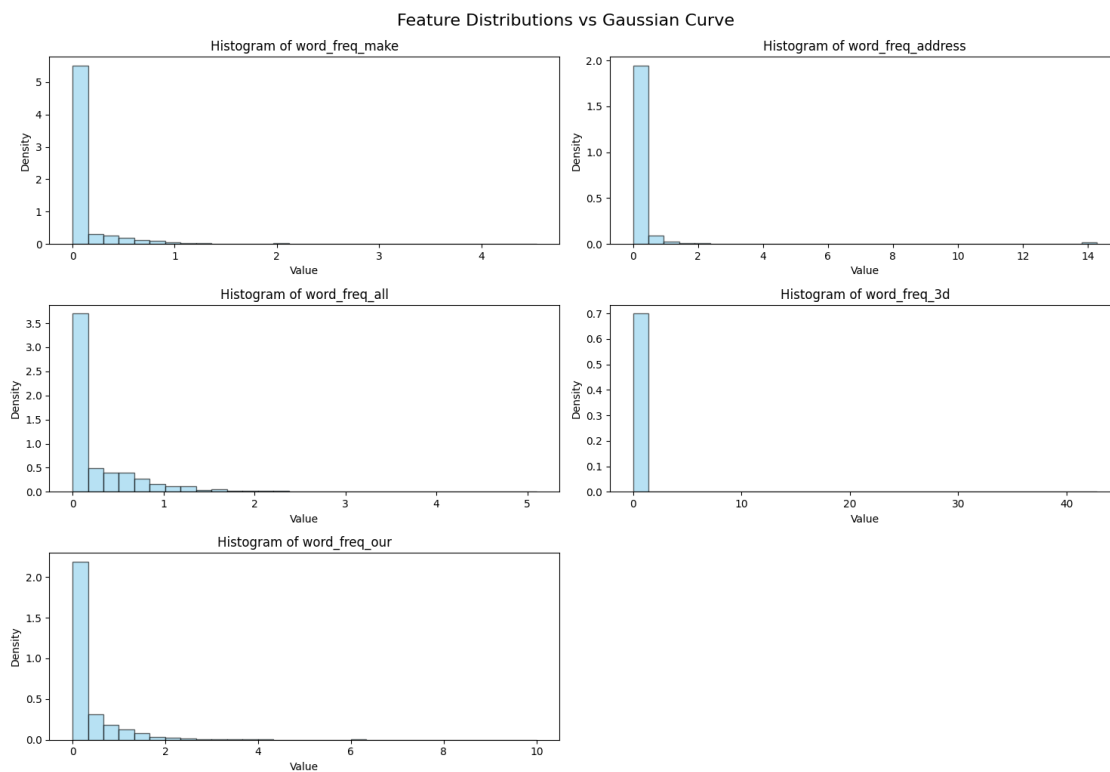
```

xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = 1 / (std * np.sqrt(2 * np.pi)) * np.exp(-(x - mu)**2 / (2 * std**2))
plt.plot(x, p, 'r', linewidth=2)'''

plt.title(f'Histogram of {feature}')
plt.xlabel('Value')
plt.ylabel('Density')

plt.tight_layout()
plt.suptitle('Feature Distributions vs Gaussian Curve', fontsize=16, y=1.02)
plt.show()

```



0.0.7 Applying min max scaling

```

[10]: scaler = MinMaxScaler()
      X_scaled = scaler.fit_transform(X)

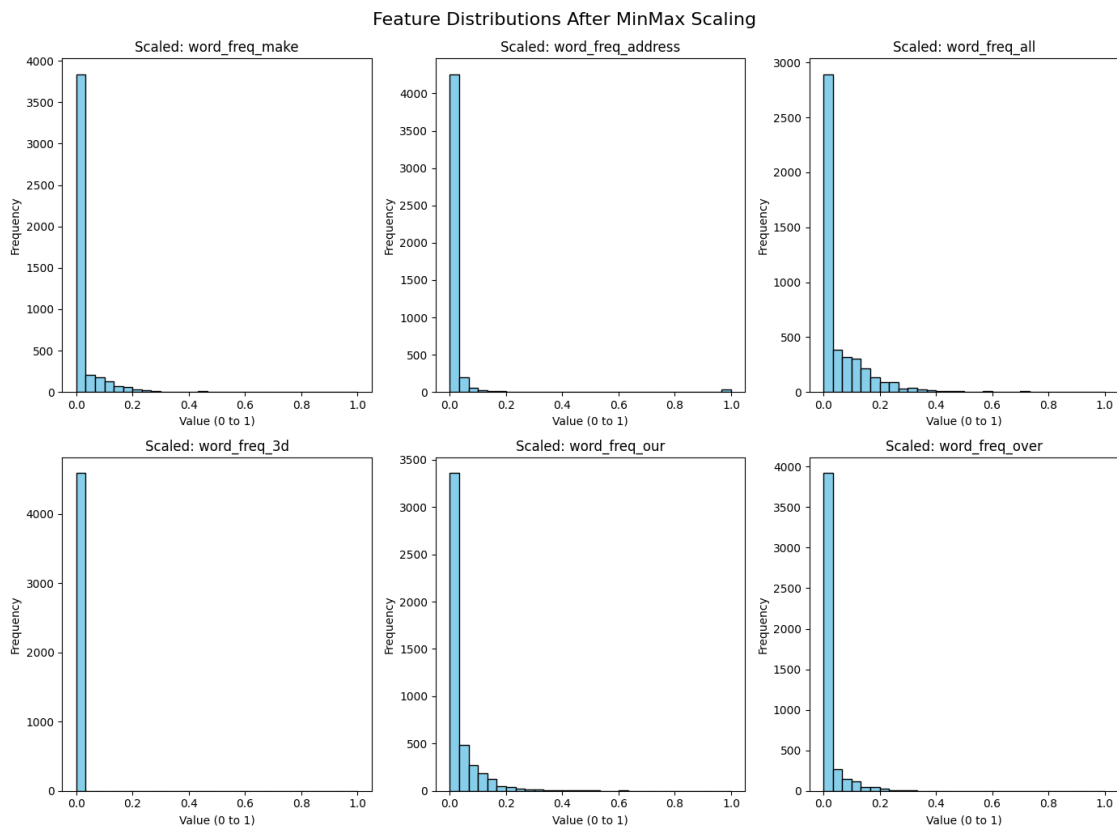
```

0.1 Plots

0.1.1 Histogram

```
[11]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))
for i in range(6): # first 6 features as an example
    plt.subplot(2, 3, i + 1)
    plt.hist(X_scaled[:, i], bins=30, color='skyblue', edgecolor='black')
    plt.title(f'Scaled: {X.columns[i]}')
    plt.xlabel('Value (0 to 1)')
    plt.ylabel('Frequency')
plt.tight_layout()
plt.suptitle('Feature Distributions After MinMax Scaling', fontsize=16, y=1.02)
plt.show()
```

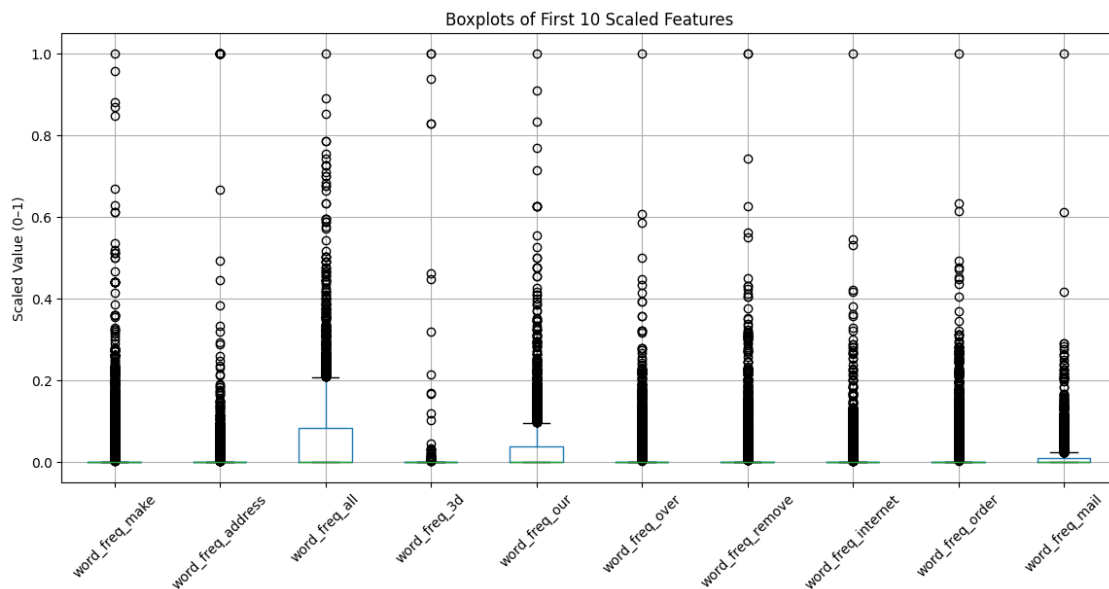


0.1.2 Boxplot

```
[12]: import pandas as pd

# Convert scaled array back to DataFrame for easier plotting
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)

plt.figure(figsize=(14, 6))
X_scaled_df.iloc[:, :10].boxplot(rot=45)
plt.title('Boxplots of First 10 Scaled Features')
plt.ylabel('Scaled Value (0-1)')
plt.grid(True)
plt.show()
```



0.1.3 Correlation HeatMap

```
[13]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Assuming X_scaled_df is your scaled DataFrame
corr_matrix = X_scaled_df.corr()

plt.figure(figsize=(18, 15)) # Bigger figure
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
```


0.1.4 Model Training

```
[14]: from sklearn.metrics import (
        accuracy_score, precision_score, recall_score,
        f1_score, confusion_matrix, roc_curve, auc,
        classification_report
    )

    # Split the data again (or reuse your earlier split)
    from sklearn.model_selection import train_test_split
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled, y, test_size=0.3, stratify=y, random_state=42
    )

[16]: def evaluate_model(name, model, X_test, y_test, results):
        y_pred = model.predict(X_test)

        # For ROC and AUC, use predict_proba
        try:
            y_proba = model.predict_proba(X_test)[: , 1]
        except:
            y_proba = y_pred # fallback if proba not available

        acc = accuracy_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred, zero_division=0)
        rec = recall_score(y_test, y_pred, zero_division=0)
        f1 = f1_score(y_test, y_pred, zero_division=0)

        # Print Confusion Matrix and Classification Report
        print(f"\n=== {name} ===")
        print("Confusion Matrix:")
        print(confusion_matrix(y_test, y_pred))
        print("\nClassification Report:")
        print(classification_report(y_test, y_pred, zero_division=0))

        # ROC and AUC
        fpr, tpr, _ = roc_curve(y_test, y_proba)
        roc_auc = auc(fpr, tpr)

        plt.plot(fpr, tpr, label=f'{name} (AUC = {roc_auc:.2f})')

        # Append results
        results.append({
            'Model': name,
            'Accuracy': acc,
            'Precision': prec,
            'Recall': rec,
```

```

        'F1 Score': f1,
        'AUC': roc_auc
    })

```

```

[17]: # Store results
results = []

# --- BernoulliNB ---
bnb = BernoulliNB()
bnb.fit(X_train, y_train)
evaluate_model("BernoulliNB", bnb, X_test, y_test, results)

# --- MultinomialNB ---
mnb = MultinomialNB()
mnb.fit(X_train, y_train)
evaluate_model("MultinomialNB", mnb, X_test, y_test, results)

# --- GaussianNB ---
gnb = GaussianNB()
gnb.fit(X_train, y_train)
evaluate_model("GaussianNB", gnb, X_test, y_test, results)

# === ROC Curve Plot ===
plt.title('ROC Curves')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

# === Final Comparison Table ===
results_df = pd.DataFrame(results)
print("\n=== Comparison Table ===")
print(results_df)

```

```

=== BernoulliNB ===
Confusion Matrix:
[[778  59]
 [ 93 451]]

```

```

Classification Report:

```

	precision	recall	f1-score	support
0.0	0.89	0.93	0.91	837
1.0	0.88	0.83	0.86	544
accuracy			0.89	1381

macro avg	0.89	0.88	0.88	1381
weighted avg	0.89	0.89	0.89	1381

=== MultinomialNB ===

Confusion Matrix:

```
[[811 26]
 [119 425]]
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.87	0.97	0.92	837
1.0	0.94	0.78	0.85	544
accuracy			0.90	1381
macro avg	0.91	0.88	0.89	1381
weighted avg	0.90	0.90	0.89	1381

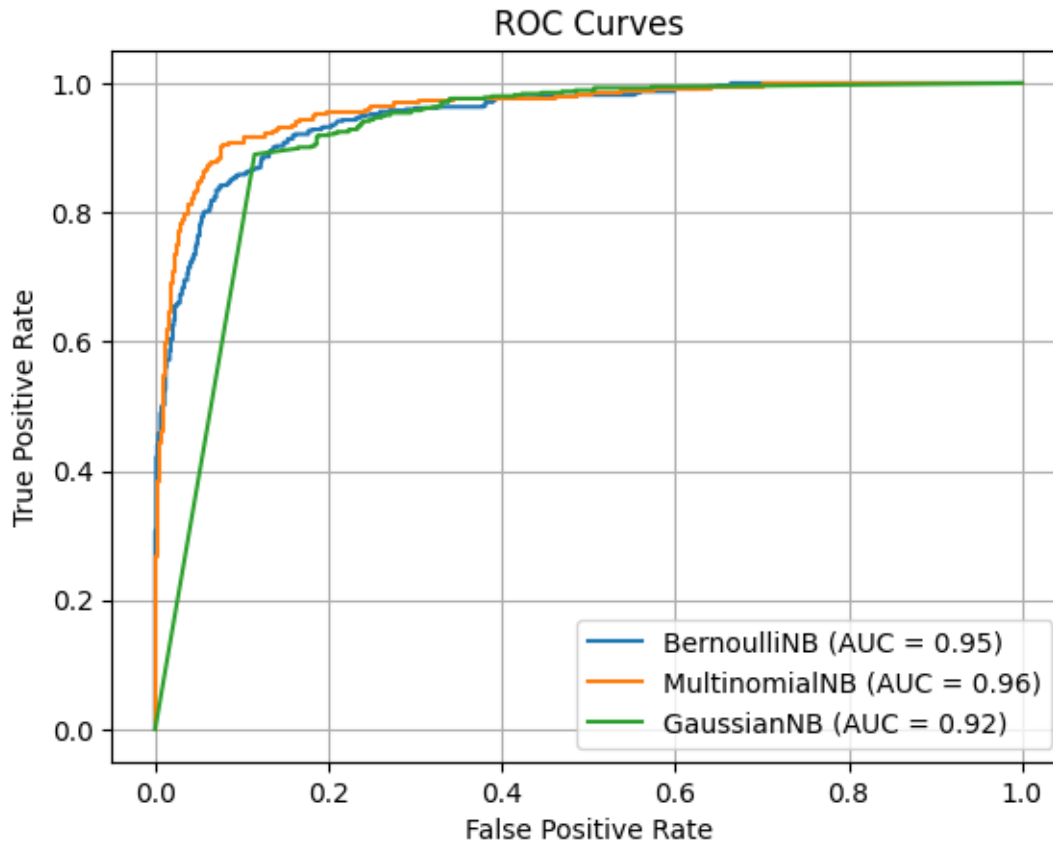
=== GaussianNB ===

Confusion Matrix:

```
[[616 221]
 [ 28 516]]
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.96	0.74	0.83	837
1.0	0.70	0.95	0.81	544
accuracy			0.82	1381
macro avg	0.83	0.84	0.82	1381
weighted avg	0.86	0.82	0.82	1381



=== Comparison Table ===

	Model	Accuracy	Precision	Recall	F1 Score	AUC
0	BernoulliNB	0.889935	0.884314	0.829044	0.855787	0.950023
1	MultinomialNB	0.895004	0.942350	0.781250	0.854271	0.960696
2	GaussianNB	0.819696	0.700136	0.948529	0.805621	0.915675

```
[20]: cv = KFold(n_splits=5, shuffle=True, random_state=42)

# Initialize Multinomial Naive Bayes
mnb = MultinomialNB()

# Perform cross-validation (accuracy as scoring)
scores = cross_val_score(mnb, X_scaled, y, cv=cv, scoring='accuracy')

# Print accuracy for each fold
print(" Multinomial Naive Bayes - 5-Fold Cross-Validation Results:")
for i, score in enumerate(scores, 1):
    print(f"Fold {i} Accuracy: {score:.4f}")
```

```
# Print average and standard deviation
print(f"\nMean Accuracy      : {scores.mean():.4f}")
print(f"Standard Deviation   : {scores.std():.4f}")
```

Multinomial Naive Bayes - 5-Fold Cross-Validation Results:

Fold 1 Accuracy:	0.8719
Fold 2 Accuracy:	0.8935
Fold 3 Accuracy:	0.8891
Fold 4 Accuracy:	0.8913
Fold 5 Accuracy:	0.8859
Mean Accuracy	: 0.8863
Standard Deviation	: 0.0077

5 KNN Code:

knn

August 5, 2025

0.0.1 Importing Required Libraries

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, confusion_matrix, \
    accuracy_score
from sklearn.impute import SimpleImputer
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import matthews_corrcoef
```

0.0.2 Loading dataset

```
[3]: df = pd.read_csv('spambase.csv')
```

0.0.3 Basic info

```
[4]: print("Dataset Info:\n", df.info())
print("\nFirst 5 rows:\n", df.head())
print("\nMissing Values:\n", df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4601 entries, 0 to 4600
```

```
Data columns (total 58 columns):
```

#	Column	Non-Null Count	Dtype
0	word_freq_make	4601 non-null	float64
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5	word_freq_over	4601 non-null	float64

6	word_freq_remove	4601 non-null	float64
7	word_freq_internet	4601 non-null	float64
8	word_freq_order	4601 non-null	float64
9	word_freq_mail	4601 non-null	float64
10	word_freq_receive	4601 non-null	float64
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20	word_freq_your	4601 non-null	float64
21	word_freq_font	4601 non-null	float64
22	word_freq_000	4601 non-null	float64
23	word_freq_money	4601 non-null	float64
24	word_freq_hp	4601 non-null	float64
25	word_freq_hpl	4601 non-null	float64
26	word_freq_george	4601 non-null	float64
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35	word_freq_technology	4601 non-null	float64
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38	word_freq_pm	4601 non-null	float64
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47	word_freq_conference	4601 non-null	float64
48	char_freq_%3B	4601 non-null	float64
49	char_freq_%28	4601 non-null	float64
50	char_freq_%5B	4601 non-null	float64
51	char_freq_%21	4601 non-null	float64
52	char_freq_%24	4601 non-null	float64
53	char_freq_%23	4601 non-null	float64

```

54 capital_run_length_average 4601 non-null float64
55 capital_run_length_longest 4601 non-null int64
56 capital_run_length_total 4601 non-null int64
57 class 4601 non-null int64

```

dtypes: float64(55), int64(3)

memory usage: 2.0 MB

Dataset Info:

None

First 5 rows:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	\
0	0.00	0.64	0.64	0.0	
1	0.21	0.28	0.50	0.0	
2	0.06	0.00	0.71	0.0	
3	0.00	0.00	0.00	0.0	
4	0.00	0.00	0.00	0.0	

	word_freq_our	word_freq_over	word_freq_remove	word_freq_internet	\
0	0.32	0.00	0.00	0.00	
1	0.14	0.28	0.21	0.07	
2	1.23	0.19	0.19	0.12	
3	0.63	0.00	0.31	0.63	
4	0.63	0.00	0.31	0.63	

	word_freq_order	word_freq_mail	...	char_freq_%3B	char_freq_%28	\
0	0.00	0.00	...	0.00	0.000	
1	0.00	0.94	...	0.00	0.132	
2	0.64	0.25	...	0.01	0.143	
3	0.31	0.63	...	0.00	0.137	
4	0.31	0.63	...	0.00	0.135	

	char_freq_%5B	char_freq_%21	char_freq_%24	char_freq_%23	\
0	0.0	0.778	0.000	0.000	
1	0.0	0.372	0.180	0.048	
2	0.0	0.276	0.184	0.010	
3	0.0	0.137	0.000	0.000	
4	0.0	0.135	0.000	0.000	

	capital_run_length_average	capital_run_length_longest	\
0	3.756	61	
1	5.114	101	
2	9.821	485	
3	3.537	40	
4	3.537	40	

	capital_run_length_total	class
0	278	1
1	1028	1

2	2259	1
3	191	1
4	191	1

[5 rows x 58 columns]

Missing Values:

word_freq_make	0
word_freq_address	0
word_freq_all	0
word_freq_3d	0
word_freq_our	0
word_freq_over	0
word_freq_remove	0
word_freq_internet	0
word_freq_order	0
word_freq_mail	0
word_freq_receive	0
word_freq_will	0
word_freq_people	0
word_freq_report	0
word_freq_addresses	0
word_freq_free	0
word_freq_business	0
word_freq_email	0
word_freq_you	0
word_freq_credit	0
word_freq_your	0
word_freq_font	0
word_freq_000	0
word_freq_money	0
word_freq_hp	0
word_freq_hpl	0
word_freq_george	0
word_freq_650	0
word_freq_lab	0
word_freq_labs	0
word_freq_telnet	0
word_freq_857	0
word_freq_data	0
word_freq_415	0
word_freq_85	0
word_freq_technology	0
word_freq_1999	0
word_freq_parts	0
word_freq_pm	0
word_freq_direct	0
word_freq_cs	0

```

word_freq_meeting      0
word_freq_original     0
word_freq_project      0
word_freq_re           0
word_freq_edu          0
word_freq_table        0
word_freq_conference   0
char_freq_%3B          0
char_freq_%28          0
char_freq_%5B          0
char_freq_%21          0
char_freq_%24          0
char_freq_%23          0
capital_run_length_average  0
capital_run_length_longest 0
capital_run_length_total  0
class                  0
dtype: int64

```

0.0.4 Handling missing values

```

[5]: imputer = SimpleImputer(strategy='mean')
     df_imputed = pd.DataFrame(imputer.fit_transform(df), columns=df.columns)

```

0.0.5 Splitting of feature and target

```

[6]: X = df_imputed.drop('class', axis=1)
     y = df_imputed['class']

```

0.0.6 Checking Distribution

```

[7]: X = df.drop('class', axis=1)  # Assuming 'class' is the target

# Choose a few features to visualize
sample_features = X.columns[:5]  # First 5 features

# Plot histograms with Gaussian curve overlay
plt.figure(figsize=(15, 10))
for i, feature in enumerate(sample_features):
    plt.subplot(3, 2, i + 1)
    data = X[feature]

    # Plot histogram
    count, bins, ignored = plt.hist(data, bins=30, density=True, alpha=0.6,
    ↪ color='skyblue', edgecolor='black')

    # Plot normal distribution curve

```

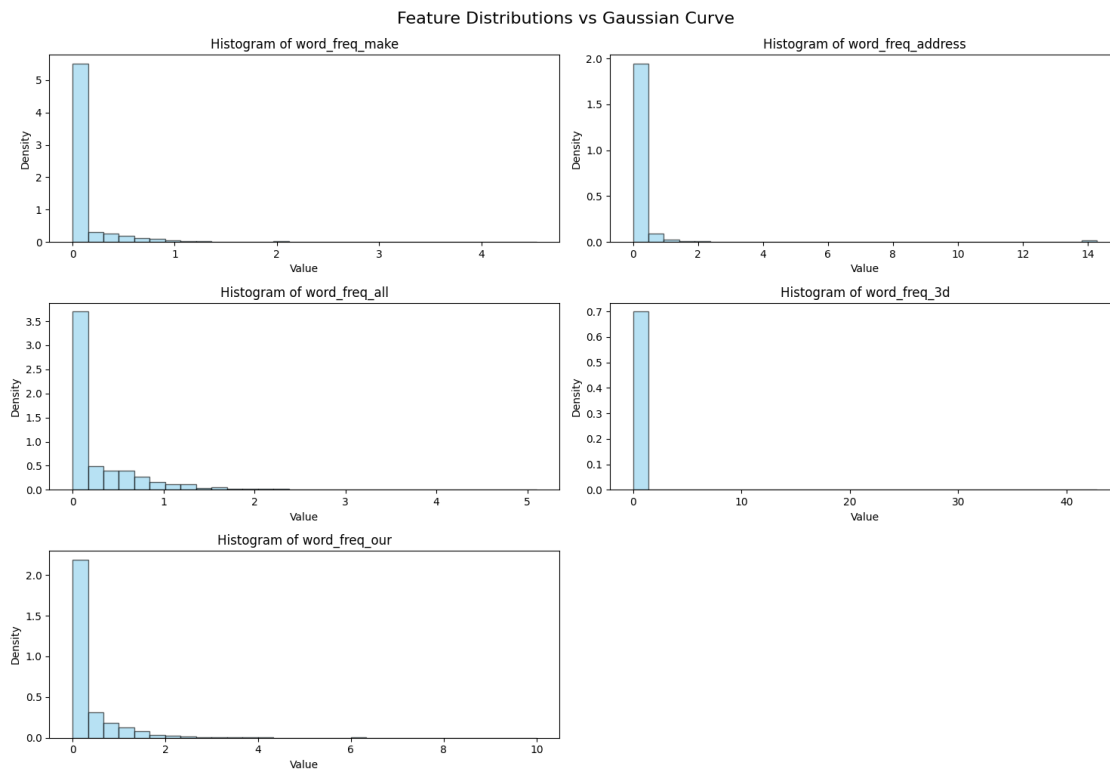
```

'''mu, std = data.mean(), data.std()
xmin, xmax = plt.xlim()
x = np.linspace(xmin, xmax, 100)
p = 1 / (std * np.sqrt(2 * np.pi)) * np.exp(-(x - mu)**2 / (2 * std**2))
plt.plot(x, p, 'r', linewidth=2)'''

plt.title(f'Histogram of {feature}')
plt.xlabel('Value')
plt.ylabel('Density')

plt.tight_layout()
plt.suptitle('Feature Distributions vs Gaussian Curve', fontsize=16, y=1.02)
plt.show()

```



0.0.7 Applying min max scaling

```

[8]: scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

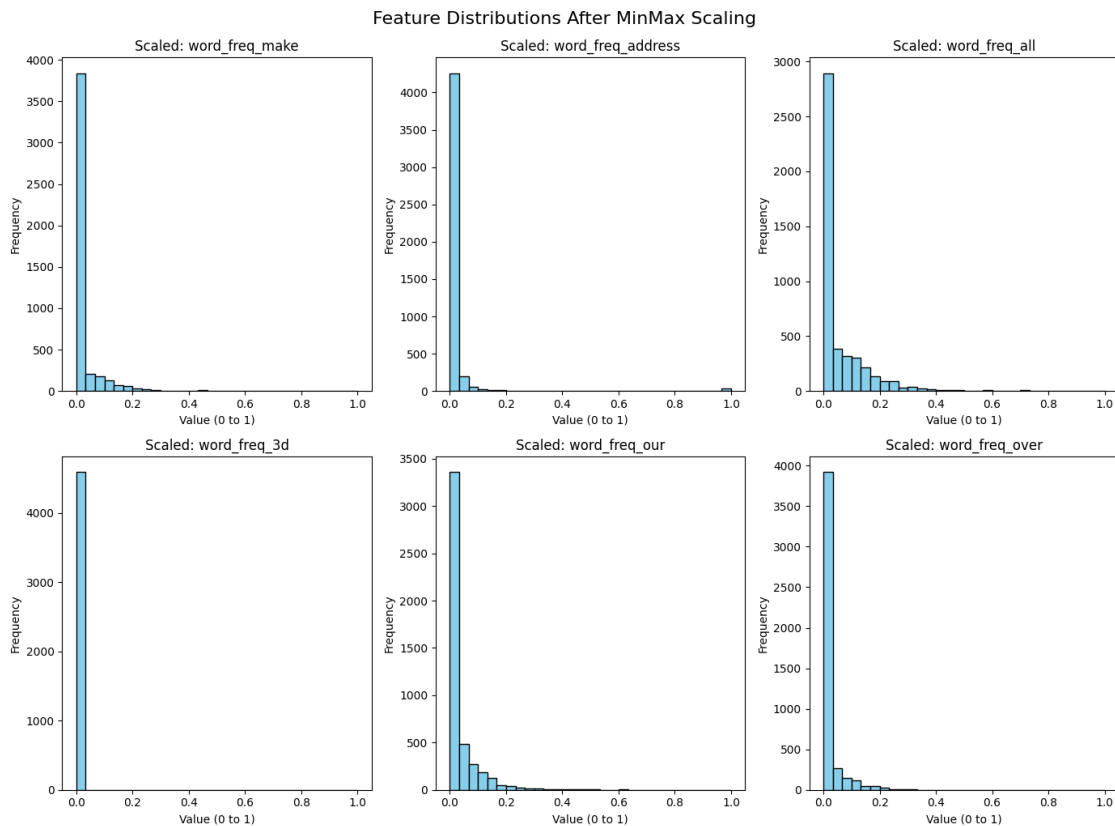
```

##Plots

0.0.8 Histogram

```
[9]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 10))
for i in range(6): # first 6 features as an example
    plt.subplot(2, 3, i + 1)
    plt.hist(X_scaled[:, i], bins=30, color='skyblue', edgecolor='black')
    plt.title(f'Scaled: {X.columns[i]}')
    plt.xlabel('Value (0 to 1)')
    plt.ylabel('Frequency')
plt.tight_layout()
plt.suptitle('Feature Distributions After MinMax Scaling', fontsize=16, y=1.02)
plt.show()
```

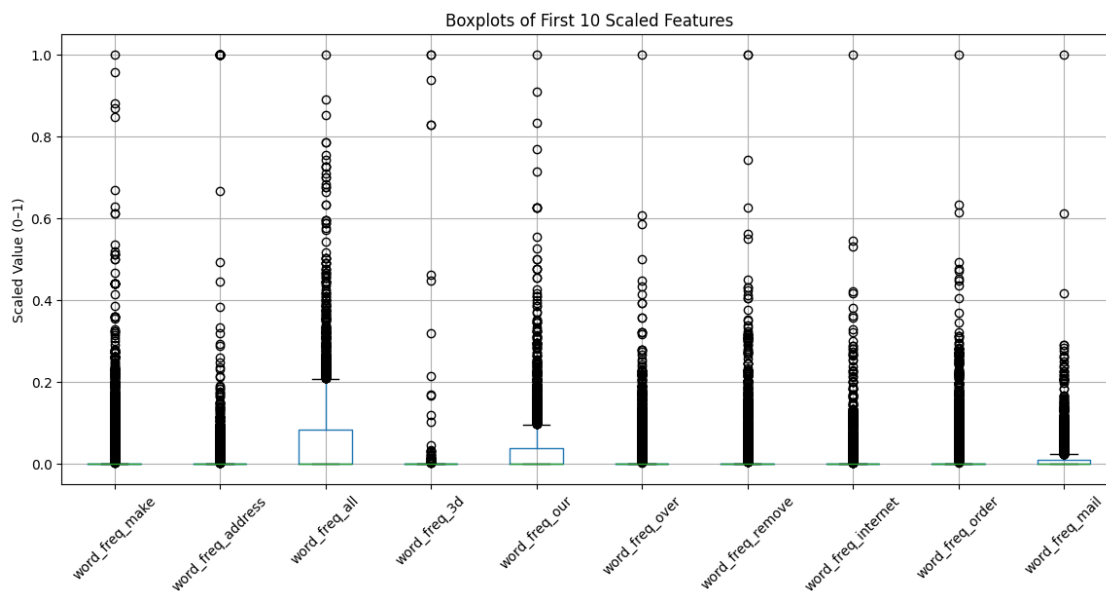


0.0.9 Boxplot

```
[10]: import pandas as pd

# Convert scaled array back to DataFrame for easier plotting
X_scaled_df = pd.DataFrame(X_scaled, columns=X.columns)
```

```
plt.figure(figsize=(14, 6))
X_scaled_df.iloc[:, :10].boxplot(rot=45)
plt.title('Boxplots of First 10 Scaled Features')
plt.ylabel('Scaled Value (0-1)')
plt.grid(True)
plt.show()
```



0.0.10 Correlation HeatMap

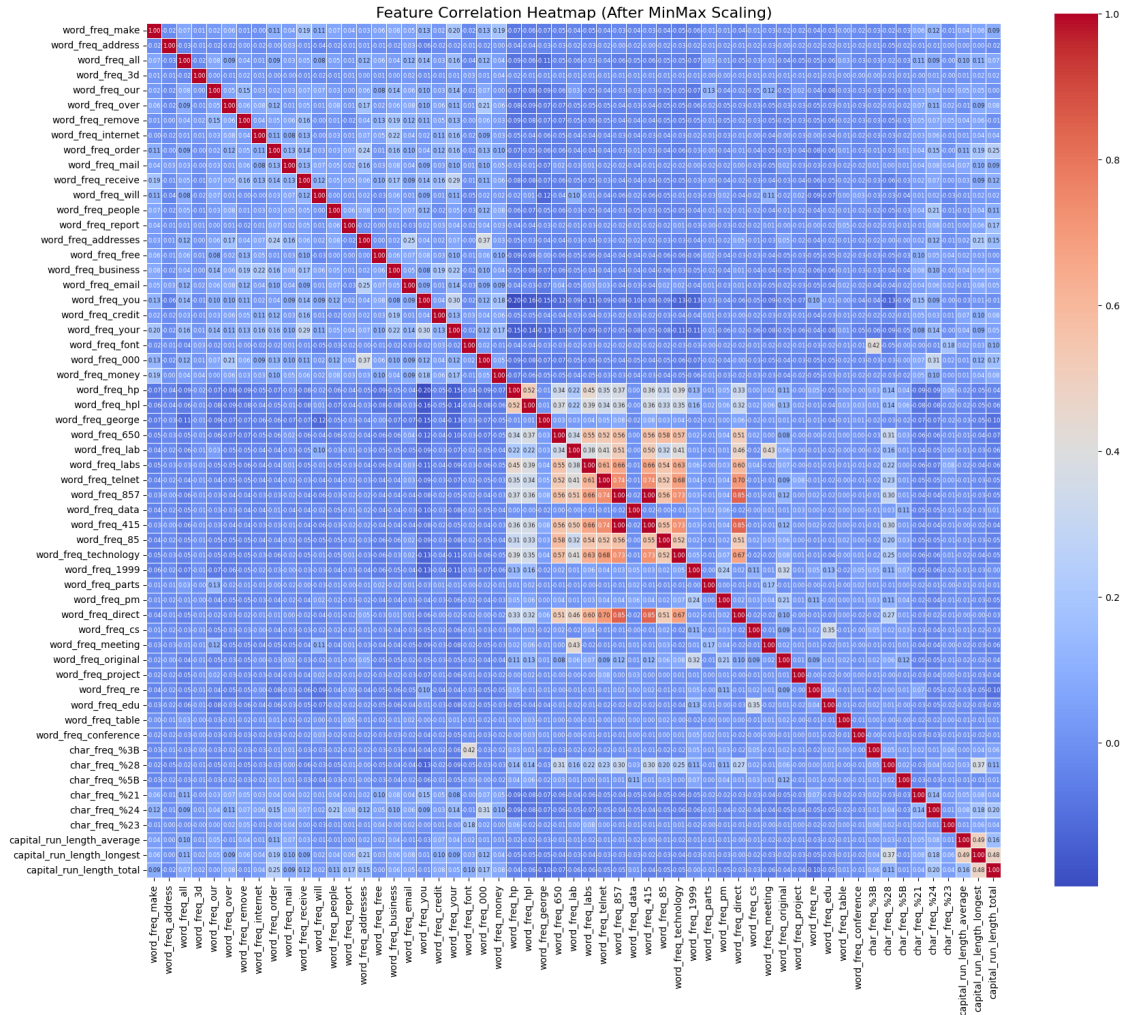
```
[11]: import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# Assuming X_scaled_df is your scaled DataFrame
corr_matrix = X_scaled_df.corr()

plt.figure(figsize=(18, 15)) # Bigger figure
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
    square=True,
    annot=True,                # Show values inside squares
    fmt=".2f",                 # Format to 2 decimal places
    linewidths=0.5,           # Thin grid lines
    annot_kws={"size": 6}     # Smaller font size
)
```



```
plt.xticks(rotation=90)
plt.yticks(rotation=0)
plt.title("Feature Correlation Heatmap (After MinMax Scaling)", fontsize=16)
plt.tight_layout()
plt.show()
```



0.0.11 Choosing best k value using k-fold cross validation

```
[12]: # Step 1: 5-Fold Stratified CV Setup
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Step 2: Try different k values
k_range = range(1, 21)
mean accuracies = []
```

```

print("Cross-Validation Accuracies for Different k:")
print("-" * 45)

for k in k_range:
    model = KNeighborsClassifier(n_neighbors=k)
    scores = cross_val_score(model, X_scaled, y, cv=cv, scoring='accuracy')
    mean_accuracies.append(scores.mean())

    print(f"k = {k:2d} -> Fold Accuracies: {np.round(scores, 4)} -> Mean_
↳Accuracy: {scores.mean():.4f}")

# Step 3: Plot Mean Accuracy vs. k
plt.figure(figsize=(10, 5))
plt.plot(k_range, mean_accuracies, marker='o', linestyle='--', color='blue')
plt.title('Mean Cross-Validation Accuracy vs. k')
plt.xlabel('Number of Neighbors (k)')
plt.ylabel('Mean Accuracy')
plt.xticks(k_range)
plt.grid(True)
plt.show()

# Step 4: Best k
best_k = k_range[np.argmax(mean_accuracies)]
print(f"\n Best k based on mean CV accuracy: {best_k}")

```

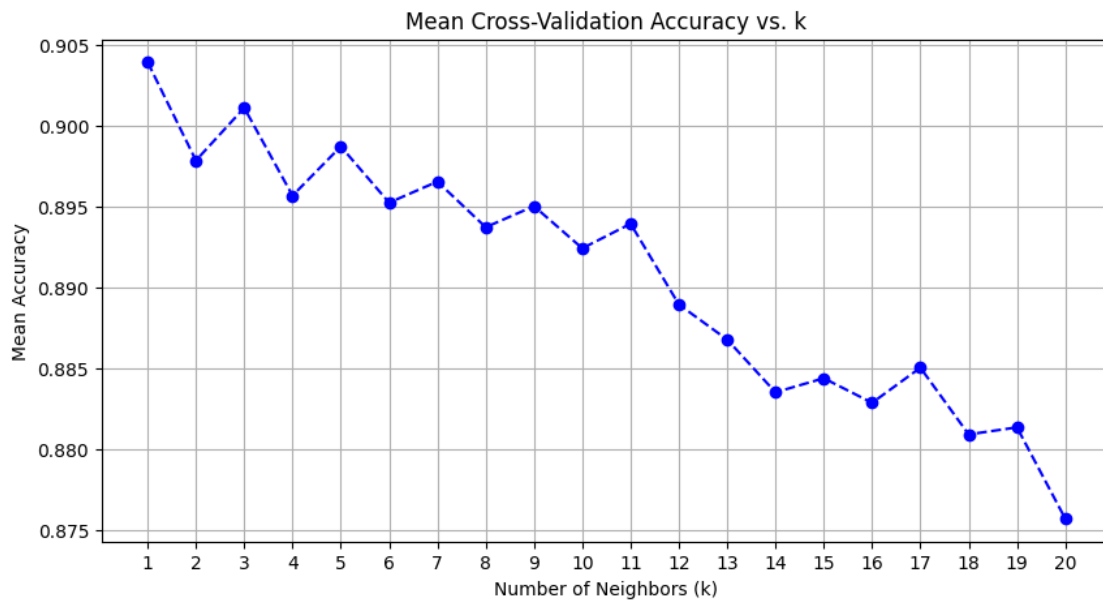
Cross-Validation Accuracies for Different k:

```

-----
k = 1 -> Fold Accuracies: [0.9034 0.9076 0.9152 0.9    0.8935] -> Mean
Accuracy: 0.9039
k = 2 -> Fold Accuracies: [0.8914 0.8924 0.9076 0.8989 0.8989] -> Mean
Accuracy: 0.8978
k = 3 -> Fold Accuracies: [0.8947 0.8989 0.9087 0.9087 0.8946] -> Mean
Accuracy: 0.9011
k = 4 -> Fold Accuracies: [0.8947 0.8902 0.9022 0.9043 0.887 ] -> Mean
Accuracy: 0.8957
k = 5 -> Fold Accuracies: [0.8969 0.8946 0.9011 0.9054 0.8957] -> Mean
Accuracy: 0.8987
k = 6 -> Fold Accuracies: [0.9001 0.8902 0.8967 0.9011 0.888 ] -> Mean
Accuracy: 0.8952
k = 7 -> Fold Accuracies: [0.899  0.8978 0.8989 0.8946 0.8924] -> Mean
Accuracy: 0.8965
k = 8 -> Fold Accuracies: [0.8914 0.8891 0.9033 0.8946 0.8902] -> Mean
Accuracy: 0.8937
k = 9 -> Fold Accuracies: [0.8936 0.8913 0.9065 0.8978 0.8859] -> Mean
Accuracy: 0.8950
k = 10 -> Fold Accuracies: [0.8947 0.888  0.9033 0.888  0.888 ] -> Mean
Accuracy: 0.8924

```

k = 11 -> Fold Accuracies: [0.8958 0.8924 0.9054 0.8957 0.8804] -> Mean Accuracy: 0.8939
 k = 12 -> Fold Accuracies: [0.8925 0.8826 0.8967 0.8924 0.8804] -> Mean Accuracy: 0.8889
 k = 13 -> Fold Accuracies: [0.8893 0.8826 0.8957 0.8935 0.8728] -> Mean Accuracy: 0.8868
 k = 14 -> Fold Accuracies: [0.8914 0.8783 0.8859 0.8891 0.8728] -> Mean Accuracy: 0.8835
 k = 15 -> Fold Accuracies: [0.8882 0.8826 0.8946 0.8913 0.8652] -> Mean Accuracy: 0.8844
 k = 16 -> Fold Accuracies: [0.8849 0.8793 0.8891 0.887 0.8739] -> Mean Accuracy: 0.8829
 k = 17 -> Fold Accuracies: [0.8893 0.887 0.8902 0.8946 0.8641] -> Mean Accuracy: 0.8850
 k = 18 -> Fold Accuracies: [0.886 0.8826 0.8902 0.8826 0.863] -> Mean Accuracy: 0.8809
 k = 19 -> Fold Accuracies: [0.8795 0.8837 0.8935 0.8902 0.8598] -> Mean Accuracy: 0.8813
 k = 20 -> Fold Accuracies: [0.8773 0.8761 0.8902 0.8837 0.8511] -> Mean Accuracy: 0.8757



Best k based on mean CV accuracy: 1

0.0.12 Model Training and Evaluation

```
[21]: from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, roc_curve, auc,
    classification_report
)

# Split the data again (or reuse your earlier split)
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, stratify=y, random_state=42
)

# Store results
knn_results = []

# Plot setup for ROC
plt.figure(figsize=(8, 6))

# Loop through different k values
for k in [1, 3, 5, 7]:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    mcc = matthews_corrcoef(y_test, y_pred)

    fpr, tpr, _ = roc_curve(y_test, y_prob)
    roc_auc = auc(fpr, tpr)

    # Plot ROC
    plt.plot(fpr, tpr, label=f'k={k} (AUC={roc_auc:.2f})')

print(f"\n KNN Evaluation for k = {k}")
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall : {rec:.4f}")
print(f"F1 Score : {f1:.4f}")
print(f"MCC : {mcc:.4f}")
print("\nClassification Report:")
```

```

print(classification_report(y_test, y_pred, zero_division=0))

knn_results.append({
    'k': k,
    'Accuracy': acc,
    'Precision': prec,
    'Recall': rec,
    'F1 Score': f1,
    'MCC': mcc,
    'AUC': roc_auc
})

# Final ROC Curve Plot
plt.title("ROC Curves for Different K Values")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.grid(True)
plt.legend()
plt.show()

# Comparison Table
knn_df = pd.DataFrame(knn_results)
print("\n Comparison Table:")
print(knn_df)

```

KNN Evaluation for k = 1

Accuracy : 0.8899
Precision: 0.8551
Recall : 0.8676
F1 Score : 0.8613
MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381

KNN Evaluation for k = 3

Accuracy : 0.8892
Precision: 0.8710
Recall : 0.8438

F1 Score : 0.8571
MCC : 0.7670

Classification Report:

	precision	recall	f1-score	support
0.0	0.90	0.92	0.91	837
1.0	0.87	0.84	0.86	544
accuracy			0.89	1381
macro avg	0.89	0.88	0.88	1381
weighted avg	0.89	0.89	0.89	1381

KNN Evaluation for k = 5

Accuracy : 0.8993
Precision: 0.8828
Recall : 0.8585
F1 Score : 0.8705
MCC : 0.7884

Classification Report:

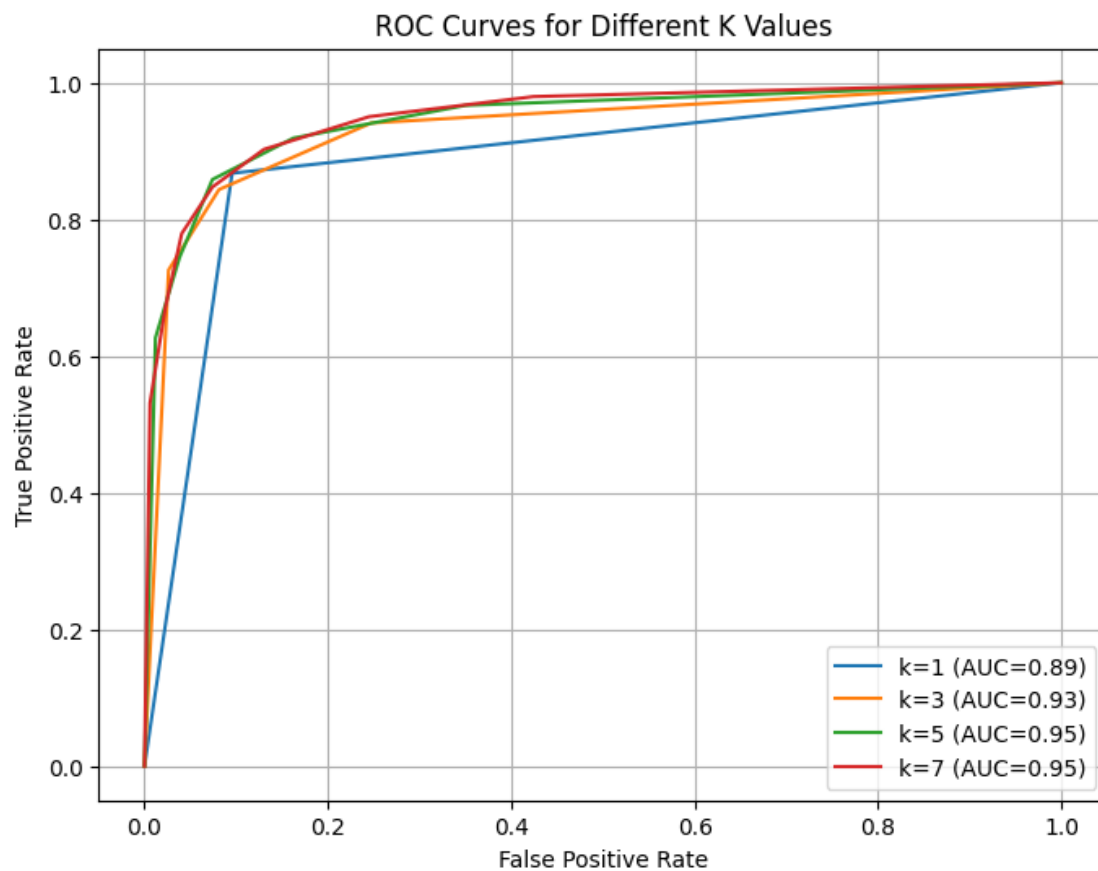
	precision	recall	f1-score	support
0.0	0.91	0.93	0.92	837
1.0	0.88	0.86	0.87	544
accuracy			0.90	1381
macro avg	0.90	0.89	0.89	1381
weighted avg	0.90	0.90	0.90	1381

KNN Evaluation for k = 7

Accuracy : 0.8950
Precision: 0.8815
Recall : 0.8474
F1 Score : 0.8641
MCC : 0.7790

Classification Report:

	precision	recall	f1-score	support
0.0	0.90	0.93	0.91	837
1.0	0.88	0.85	0.86	544
accuracy			0.90	1381
macro avg	0.89	0.89	0.89	1381
weighted avg	0.89	0.90	0.89	1381



Comparison Table:

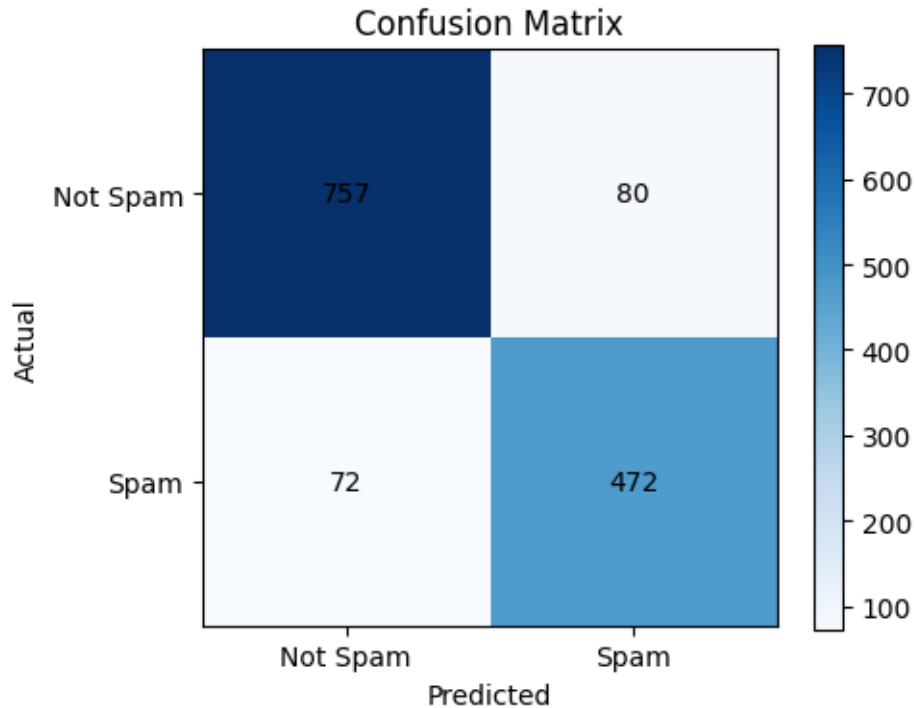
	k	Accuracy	Precision	Recall	F1 Score	MCC	AUC
0	1	0.889935	0.855072	0.867647	0.861314	0.770142	0.886034
1	3	0.889211	0.870968	0.843750	0.857143	0.766959	0.931442
2	5	0.899348	0.882798	0.858456	0.870457	0.788392	0.945145
3	7	0.895004	0.881453	0.847426	0.864105	0.779014	0.950240

```
[15]: conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(5, 4))
plt.imshow(conf_matrix, cmap='Blues')
plt.title('Confusion Matrix')
plt.colorbar()
plt.xticks([0, 1], ['Not Spam', 'Spam'])
plt.yticks([0, 1], ['Not Spam', 'Spam'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
for i in range(2):
```

```

    for j in range(2):
        plt.text(j, i, conf_matrix[i, j], ha='center', va='center',
        ↪color='black')
plt.tight_layout()
plt.show()

```



0.0.13 KD Tree and Ball Tree

```

[17]: # Use KDTree
model_kd = KNeighborsClassifier(n_neighbors=best_k, algorithm='kd_tree')
model_kd.fit(X_train, y_train)

# Use BallTree
model_ball = KNeighborsClassifier(n_neighbors=best_k, algorithm='ball_tree')
model_ball.fit(X_train, y_train)

```

```

[17]: KNeighborsClassifier(algorithm='ball_tree', n_neighbors=1)

```

```

[18]: from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, roc_curve, auc,
    classification_report, matthews_corrcoef
)

```



```

)
import matplotlib.pyplot as plt

# Helper function to evaluate a model
def evaluate_knn_model(name, model, X_train, X_test, y_train, y_test):
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    mcc = matthews_corrcoef(y_test, y_pred)
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    roc_auc = auc(fpr, tpr)

    print(f"\n {name} Performance Metrics:")
    print(f"Accuracy : {acc:.4f}")
    print(f"Precision: {prec:.4f}")
    print(f"Recall    : {rec:.4f}")
    print(f"F1 Score  : {f1:.4f}")
    print(f"MCC      : {mcc:.4f}")
    print("\nClassification Report:\n", classification_report(y_test, y_pred))

    # Confusion Matrix
    conf = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(5, 4))
    plt.imshow(conf, cmap='Blues')
    plt.title(f'{name} Confusion Matrix')
    plt.colorbar()
    plt.xticks([0, 1], ['Not Spam', 'Spam'])
    plt.yticks([0, 1], ['Not Spam', 'Spam'])
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    for i in range(2):
        for j in range(2):
            plt.text(j, i, conf[i, j], ha='center', va='center')
    plt.tight_layout()
    plt.show()

    # ROC Curve
    plt.figure(figsize=(6, 5))
    plt.plot(fpr, tpr, label=f'{name} (AUC = {roc_auc:.4f})')
    plt.plot([0, 1], [0, 1], 'k--', linewidth=1)
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')

```

```

plt.title(f'{name} ROC Curve')
plt.legend()
plt.grid(True)
plt.show()

# Train-test split assumed done:
# X_train, X_test, y_train, y_test already available
# best_k is selected

# Evaluate KDTree
knn_kd = KNeighborsClassifier(n_neighbors=best_k, algorithm='kd_tree')
evaluate_knn_model("KDTree", knn_kd, X_train, X_test, y_train, y_test)

# Evaluate BallTree
knn_ball = KNeighborsClassifier(n_neighbors=best_k, algorithm='ball_tree')
evaluate_knn_model("BallTree", knn_ball, X_train, X_test, y_train, y_test)

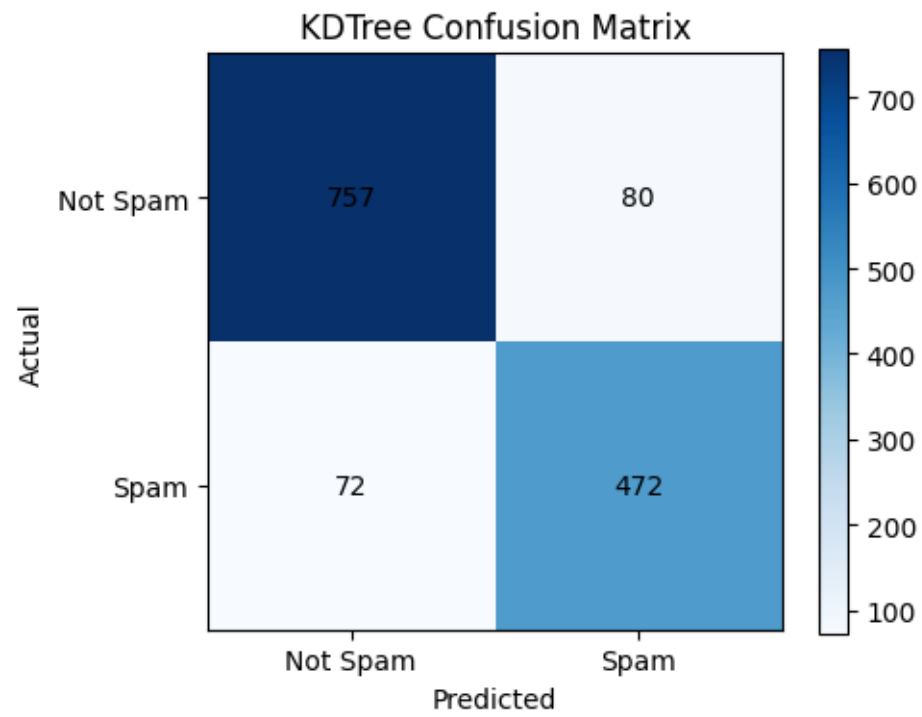
```

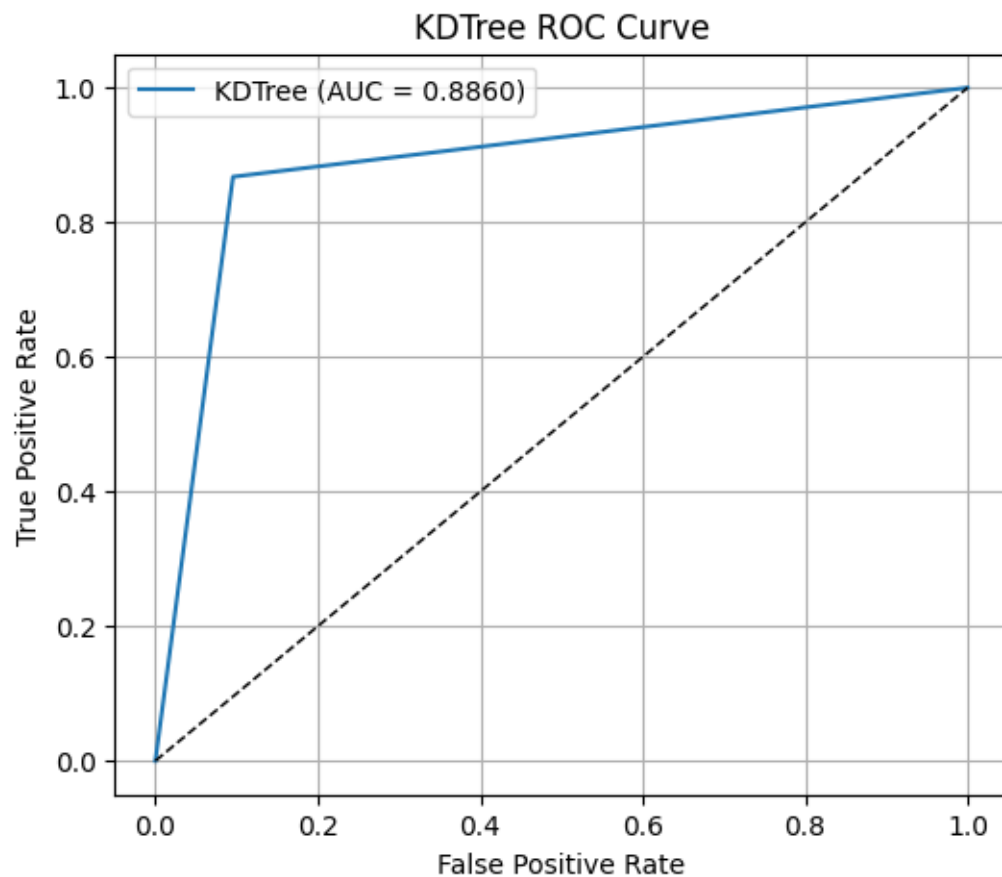
KDTree Performance Metrics:

Accuracy : 0.8899
 Precision: 0.8551
 Recall : 0.8676
 F1 Score : 0.8613
 MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381





BallTree Performance Metrics:

Accuracy : 0.8899

Precision: 0.8551

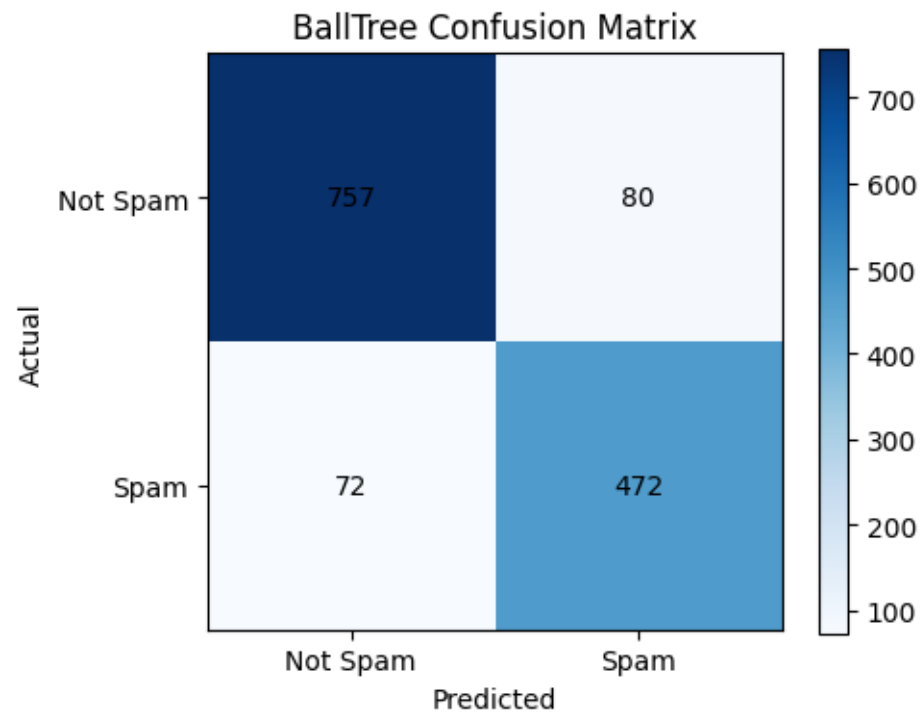
Recall : 0.8676

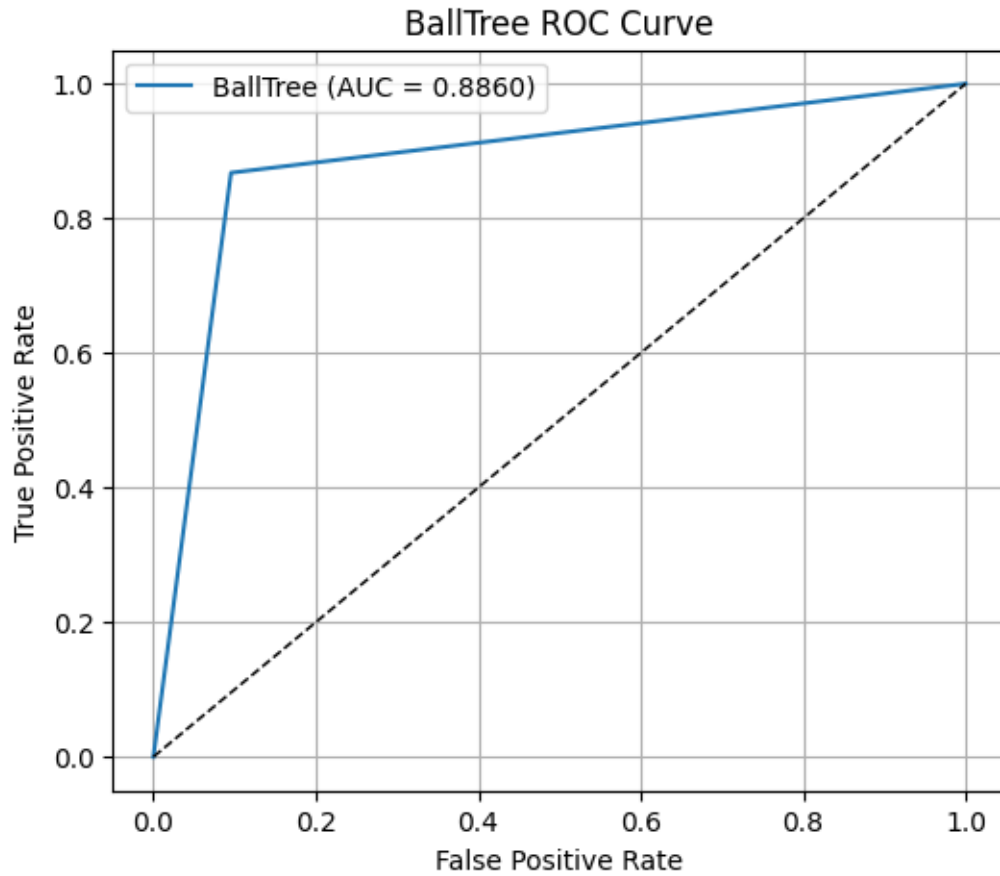
F1 Score : 0.8613

MCC : 0.7701

Classification Report:

	precision	recall	f1-score	support
0.0	0.91	0.90	0.91	837
1.0	0.86	0.87	0.86	544
accuracy			0.89	1381
macro avg	0.88	0.89	0.89	1381
weighted avg	0.89	0.89	0.89	1381





```
[26]: import time

for algo in ['kd_tree', 'ball_tree', 'brute']:
    model = KNeighborsClassifier(n_neighbors=best_k, algorithm=algo)
    start = time.time()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    duration = time.time() - start
    acc = accuracy_score(y_test, y_pred)
    print(f"{algo:10} → Accuracy: {acc:.4f}, Time: {duration:.4f} sec")
```

```
kd_tree    → Accuracy: 0.8899, Time: 0.4470 sec
ball_tree  → Accuracy: 0.8899, Time: 0.4058 sec
brute      → Accuracy: 0.8899, Time: 0.0192 sec
```

6 Comparison Tables

Table 1: Performance Comparison of Naïve Bayes Variants

Model	Accuracy	Precision	Recall	F1 Score
BernoulliNB	0.8899	0.8843	0.8290	0.8558
MultinomialNB	0.8950	0.9424	0.7813	0.8543
GaussianNB	0.8197	0.7001	0.9485	0.8056

Table 2: KNN Performance for Different k Values

k	Accuracy	Precision	Recall	F1 Score
1	0.8899	0.8551	0.8676	0.8613
3	0.8892	0.8710	0.8438	0.8571
5	0.8993	0.8828	0.8585	0.8705
7	0.8950	0.8815	0.8474	0.8641

Table 3: KNN Comparison: KDTree vs BallTree

Metric	KDTree	BallTree
Accuracy	0.8899	0.8899
Precision	0.8551	0.8551
Recall	0.8676	0.8676
F1 Score	0.8613	0.8613
Training Time (s)	0.4470	0.4058

Table 4: Cross-Validation Scores for Each Model (K = 5)

Fold	Naïve Bayes Accuracy	KNN Accuracy (k=1)	SVM Accuracy
Fold 1	0.8719	0.9034	—
Fold 2	0.8935	0.9076	—
Fold 3	0.8891	0.9152	—
Fold 4	0.8913	0.9000	—
Fold 5	0.8859	0.8935	—
Average	0.8863	0.9039	—

7 Observation:

- KNN (k=1) consistently achieved higher accuracy across all 5 folds compared to Multinomial Naïve Bayes, with an average accuracy of 90.39 percentage vs 88.63 percentage.
- Naïve Bayes showed slightly more stable performance across folds, whereas KNN had slightly higher variance but better peak performance (e.g., Fold 3 with 91.52 percentage).
- KNN's superior accuracy suggests that instance-based learning works better for this dataset, possibly due to well-separated class boundaries that KNN can capture using distance metrics.

8 Conclusion:

- KNN (k=1) outperformed Multinomial Naïve Bayes in terms of average accuracy in 5-fold cross-validation, making it the better choice for this dataset.
- However, Naïve Bayes remains a strong baseline due to its simplicity, speed, and competitive accuracy, especially for high-dimensional or text data.

GitHub Link